HC for III set up = RL of G + BS of G = 21.170 + 0.900 = 22.070
RL of H = HC for III set up – IS of H = 22.070 - 2.100 = 19.970
HC for IV set up = RL of H + BS of H = 19.970 + 1.560 = 21.530
RL of I = HC for IV set up – IS of I = 21.530 - 2.200 = 19.330

Arithmetic check

Σ BS ~ Σ FS = first RL ~ last RL
4.780 ~ 10.450 = 25.000 ~ 19.330
5.670 = 5.670

Hence checked

Type (II)Rise and fall method

Station	BS	IS	FS	Rise (+)	Fall (–)	RL	Remarks
A	1.120					25.000	BM
B		1.650			0.530	24.470	
C		1.750			0.100	24.370	
D	1.200		2.900		1.150	23.220	CP1
E		1.660			0.460	22.760	
F		2.520			0.860	21.900	
G	0.900		3.250		0.730	21.170	CP2
H	1.560		2.100		1.200	19.970	
I			2.200		0.640	19.330	
Total	4.780		10.450		5.670		

Calculation

1. Level difference between points A and B = 1.120 - 1.650 = -0.530
2. Level difference between points B and C = 1.650 - 1.750 = -0.100
3. Level difference between points C and D = 1.750 – 2.900 = -0.150
4. Level difference between points D and E = 1.200 - 1.660 = -0.460
5. Level difference between points E and F = 1.660 – 2.520 = -0.860
6. Level difference between points F and G = 2.520 – 3.250 = -0.730
7. Level difference between points G and H = 0.900 – 2.100 = -1.200
8. Level difference between points H and I = 1.560 – 2.200 = -0.640

Arithmetic check

Σ BS ~ Σ FS = Σ Rise ~ Σ Fall = first RL ~ last RL
4.780 ~ 10.450 = 0 ~ 5.670 =25.000 ~ 19.330
5.670 = 5.670 = 5.670

Hence checked

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